

A Survey on the Perceptual Visibility of Overlaid Content on Optical See-Through Augmented Reality

光学シースルー拡張現実における重畳コンテンツの
知覚的視認性に関する調査

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2. Studies on the Perceptual Visibility of OST-AR
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- f. Scope of this Survey

2. Studies on the Perceptual Visibility of OST-AR

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4. Conclusion

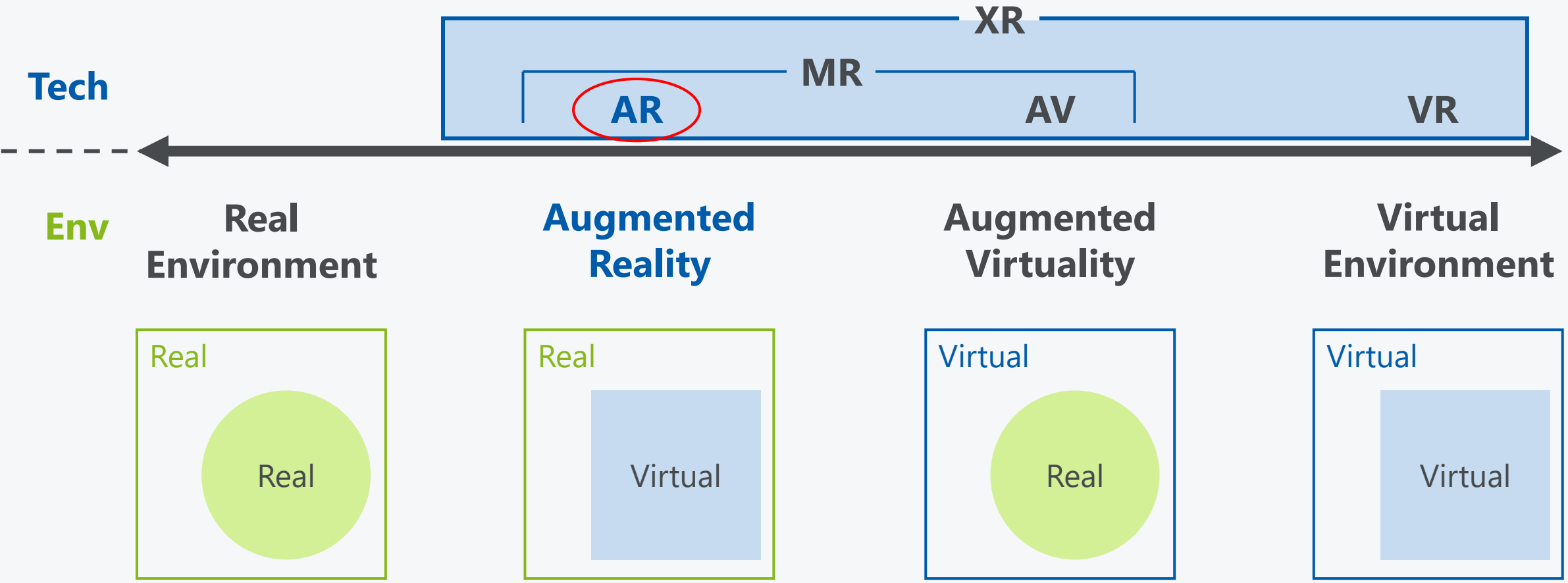
What is AR?

Virtual in Real

<Virtual Continuum[1]>



Example of AR[2]

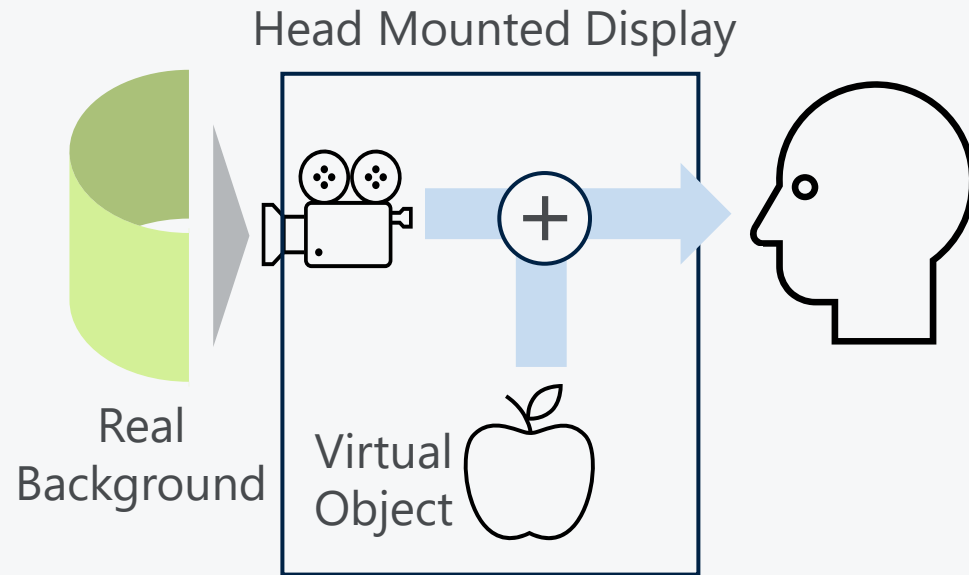


[1] F. K. Paul MILGRAM, "A taxonomy of mixed reality visual displays," 1994

[2] Y. Zhang, R. Wang, Y. Peng, W. Hua, and H. Bao, "Color Contrast Enhanced Rendering for Optical See Through Head-Mounted Displays," 2022

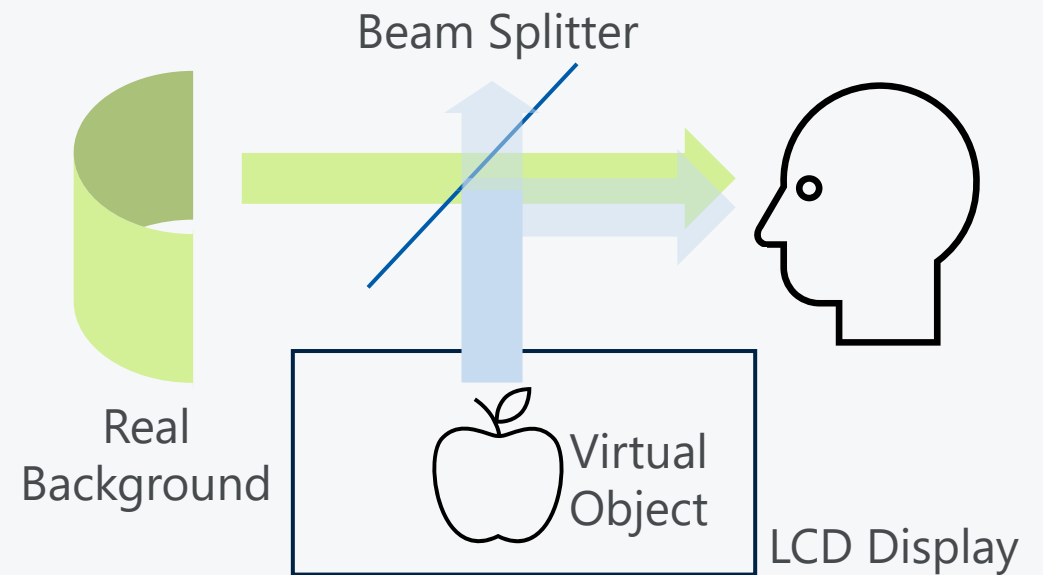
What is “OST”-AR?^[1]

Video See-Through AR



- Wide field of view
- Relatively controlled rendering
- ✗ Resolution loss & latency of BG

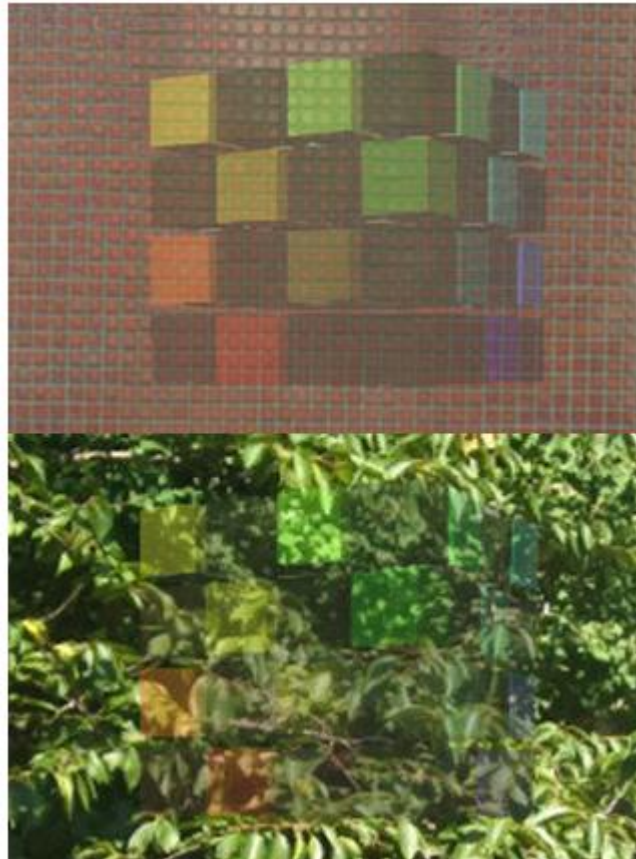
Optical See-Through AR



- Direct perception of BG
- ✗ Reduction in visibility
- ✗ Inaccurate perception

Issue of OST-AR

Reduction of Visibility Based on Background



[1]



[2]

[1] T. Fukiage, T. Oishi, and K. Ikeuchi, "Visibility-based blending for real-time applications," 2014

[2] H. Liu, et al. "Saliency difference based objective evaluation method for a superimposed screen of the hud with various background," 2019

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2. Studies on the Perceptual Visibility of OST-AR

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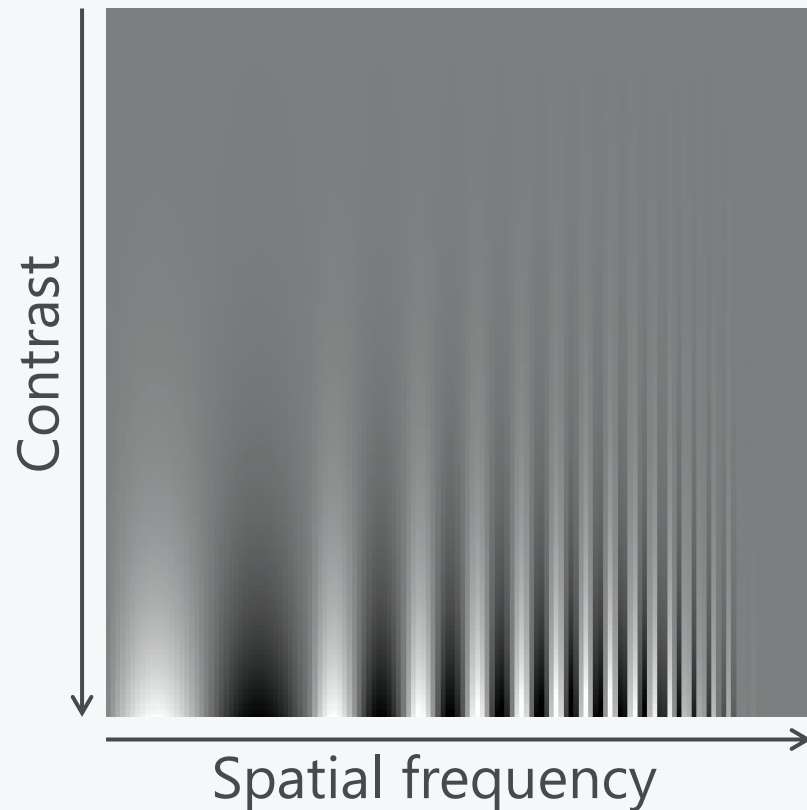
4. Conclusion

What is "Perceptual Visibility"? (1): Contrast

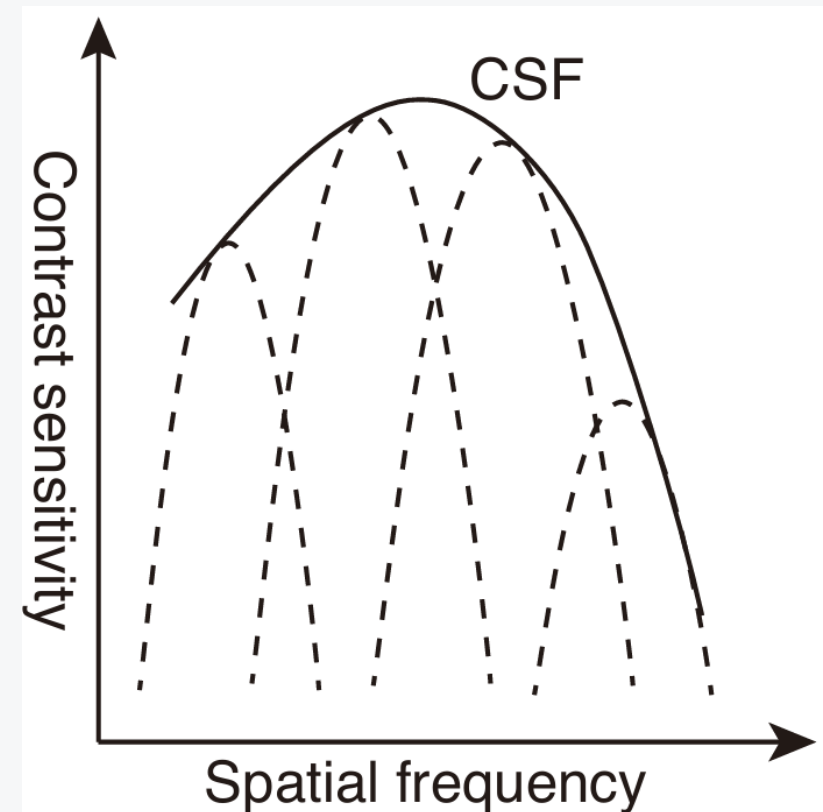
Perceptual Visibility Depends on Contrast

High contrast → High visibility

Dependence on spatial frequency



Contrast sensitivity for a visual stimulus depends on its spatial frequency[1]



Schematic illustration of the contrast sensitivity function[1]

What is "Perceptual Visibility"?(2): Interpretations

Subjectivity in Direct Visibility Assessment → Diverse Interpretations

Depend on : Criteria, Focus factors, Experimental design

Study	Focus Factor	Interpretations
Yoon et al. [1]	Color (Hue)	Hue difference against background
Kim & Lee [2]	Luminance & Illuminance	Appropriateness of brightness
Hussain & Park [3]	User Interface	Task-based visibility

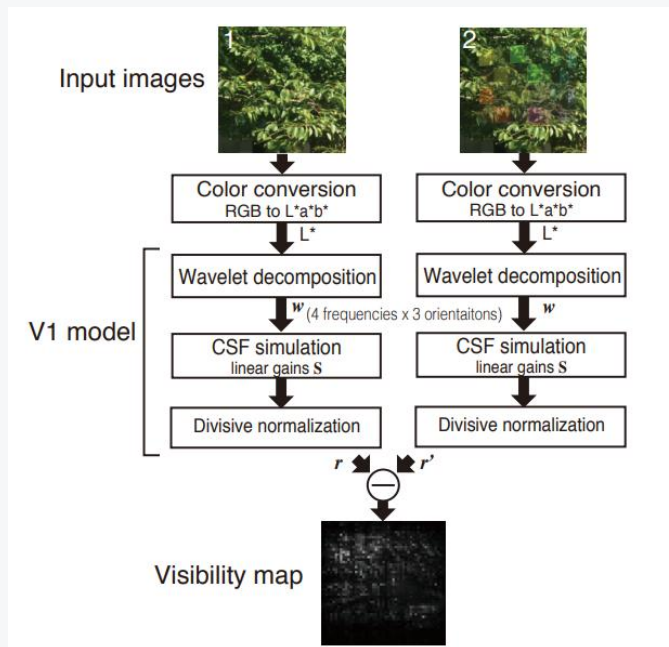
[1] H. J. Yoon, Y. Park, and H.-Y. Jung, "Background scene dominant color based visibility enhancement of head-up display," 2017
[2] J.-S. Kim and S.-W. Lee, "Study on how to improve visibility of transparent display for augmented reality under various environment conditions," 2020
[3] M. Hussain and J. Park, "Effect of transparency levels and real-world backgrounds on the user interface in augmented reality environments," 2024

Applications (1): Image Quality Assessment(IQA)

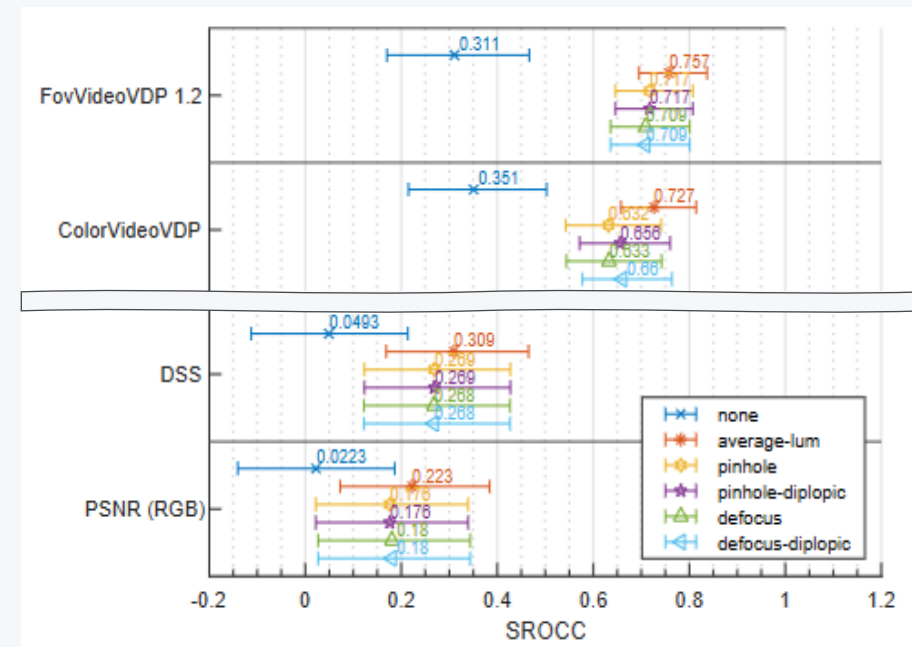
Perceptual Visibility $\Leftrightarrow$ Perceptual Image Quality

Visibility prediction by IQA Model[1]

OST-AR IQA[2]



Schematic of the visibility model[1]



Performance of the image/video quality metric[2]

→ Limitations of existing IQA Model in OST-AR[2][3]

[1] T. Fukiage, T. Oishi, and K. Ikeuchi, "Visibility-based blending for real-time applications," 2014

[2] A. Chapiro, D. Kim, Y. Asano, and R. K. Mantiuk, "Ar-david: Augmented reality display artifact video dataset," 2024

[3] T. Fukiage and T. Oishi, "A content-adaptive visibility predictor for perceptually optimized image blending," 2023

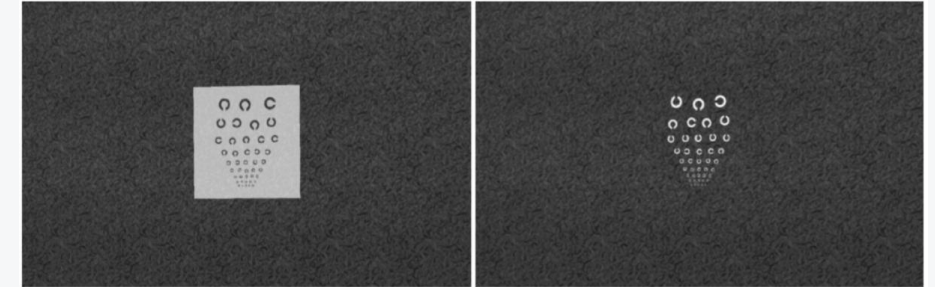
Applications (2): UX & Safety

UX: Dark Mode Achieves High Visibility and Preference in OST-AR[1]

Dark Mode achieves high visibility and preferences



Traditional screens



Light mode and dark mode in OST-AR[1]

Safety: Color Interference Reduces Contrast[2]

Many warm light sources in the driving environment



Visibility of Red color is often reduced



The ambient light attenuated the saliency of the projected information on the HUD[2]

[1] A. Erickson et al., "An extended analysis on the benefits of dark mode user interfaces in optical see through head-mounted displays," 2021

[2] H. Liu, et al. "Saliency difference based objective evaluation method for a superimposed screen of the hud with various background," 2019

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Scope of this Survey

Perceptual Visibility of Virtual Content on OST-AR

Topic Factors :

(1) Color (2) Brightness (3) Transparency

Review Structure :

Basic Research	Applied Research
Effect on perceptual visibility	Visibility prediction & enhancement

Additional Focus : **Visibility Predictor**

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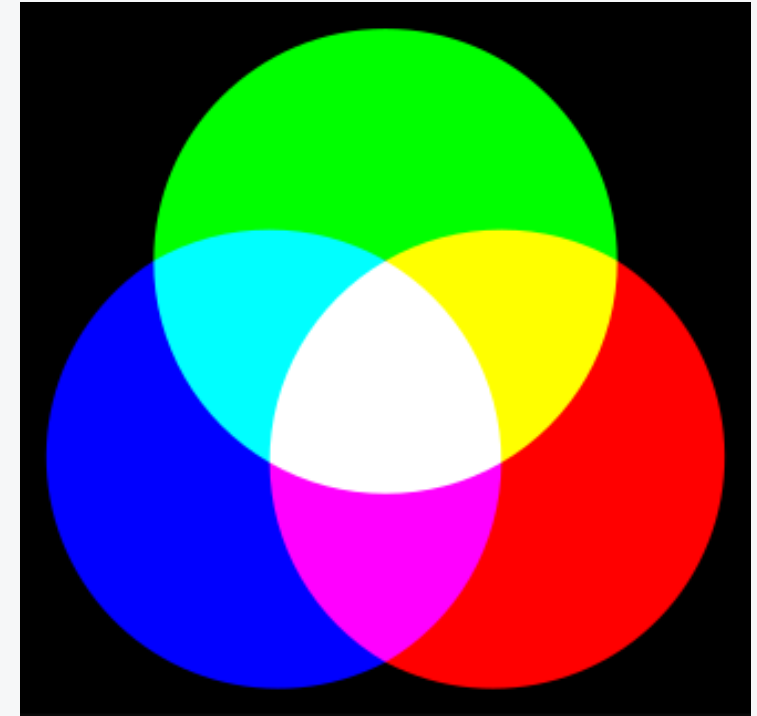
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Color Blending and Visibility

Perceived Color in OST-AR = Additive Mixture

Challenges due to color blending

- Inaccurate color perception
- Reduction in visibility



Additive Mixture(Wikipedia)

Research Outline

Paper1: Engineering Study of Color Blending in OST-AR

→Understanding about Color Blending in OST-AR

Paper2: Effects of Color Blending on OST-AR

→Understanding about effect on Perceptual Visibility

Paper3: Visibility Enhanced Rendering based on Color Contrast

→Introducing Visibility Enhancing Method in this Topic

Paper3-1: Objective and Method

Objective: Understanding and Quantifying Color Blending in OST-AR

Method: Construction of an Optical Testbed Emulating Color Blending

Testbed specifications

Lighting: D65 (Standard Daylight), 800-1000 lux

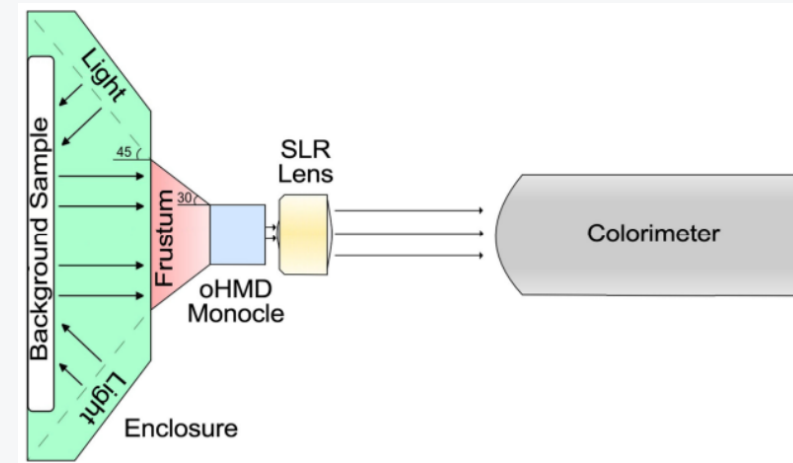
Sensor: Colorimeter

Backgrounds: 6 types (4 textures and White, None)

Virtual Stimuli: 27 RGB Combinations

Analysis of color shift

in CIE1976L*u*v* space



Schematic of experimental testbed



Color-matched sample

Paper3-1: Results and Future Work

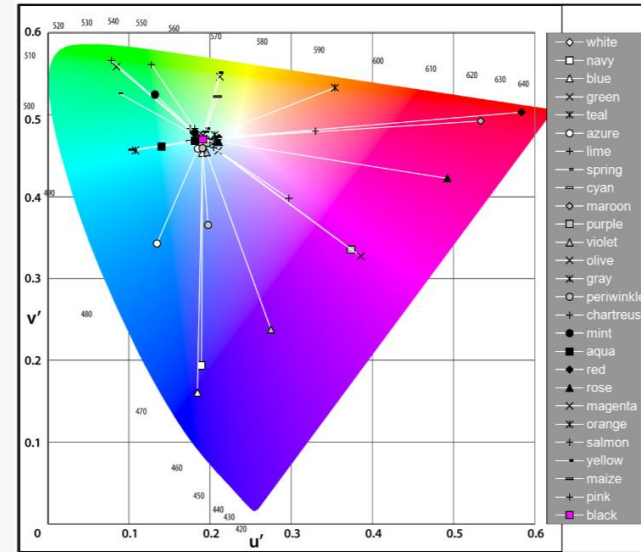
Results: Convergence to the White Point and Linear Hue Shift

No BG → White BG: Lightness effect

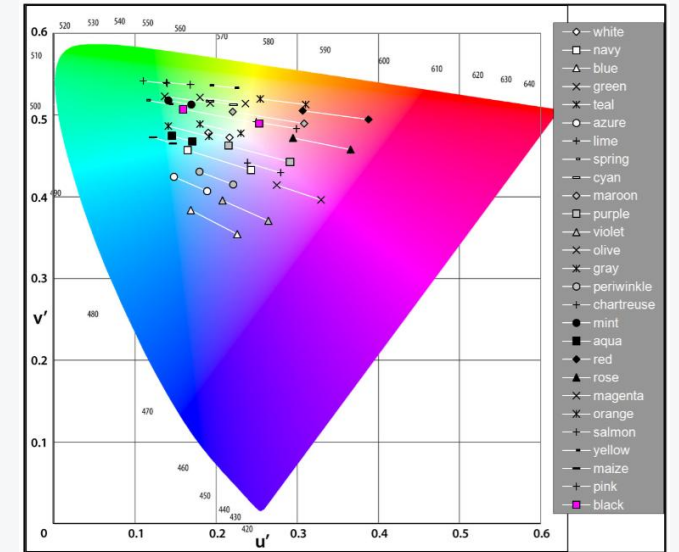
Shift toward white point (wash out)

Brick → Foliage: Color effect

Linear shift toward different hue



Chromatic changes between the no-background condition and the white background



Chromatic changes between the brick background and the foliage background

Future Work: Application and Refinement of Findings

Additional data analysis, User-based perceptual study, Testbed expansion

→ Adaptive model construction

Paper3-2: Objective and Method

Objective:

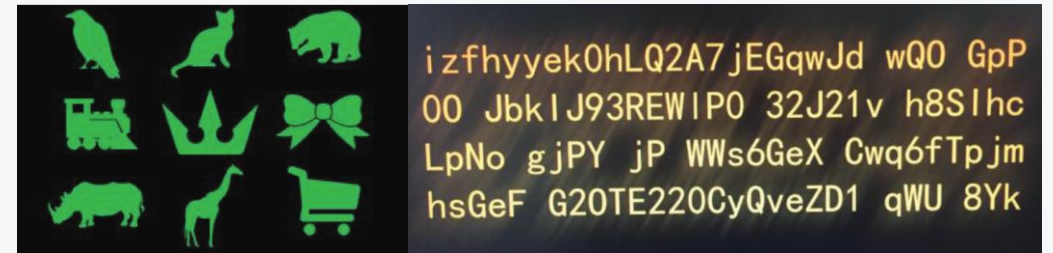
Investigation of Robust Colors for OST-HUDs in Driving Environment

Robustness: (1) low performance variability (2) retained color semantics

Method: 1. Visual Search Tasks → 2. Color Recognition Tasks

Visual Search: Text and symbol search

Color Recognition: Color matching and naming

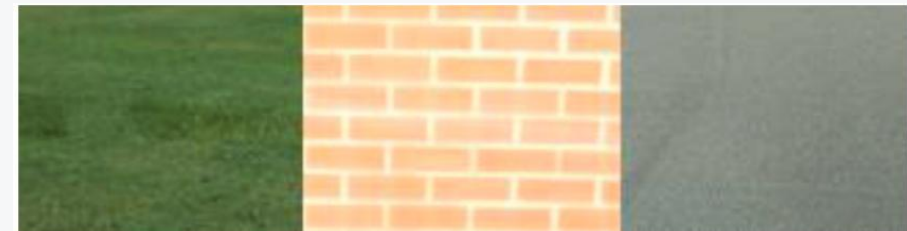


Example of symbols and texts for visual search

Location: Outdoor (500-2000 lux)

Background: 3 types (Grass, Brick, Pavement)

UI color: 8 types



Backgrounds

Paper3-2: Results and Discussion

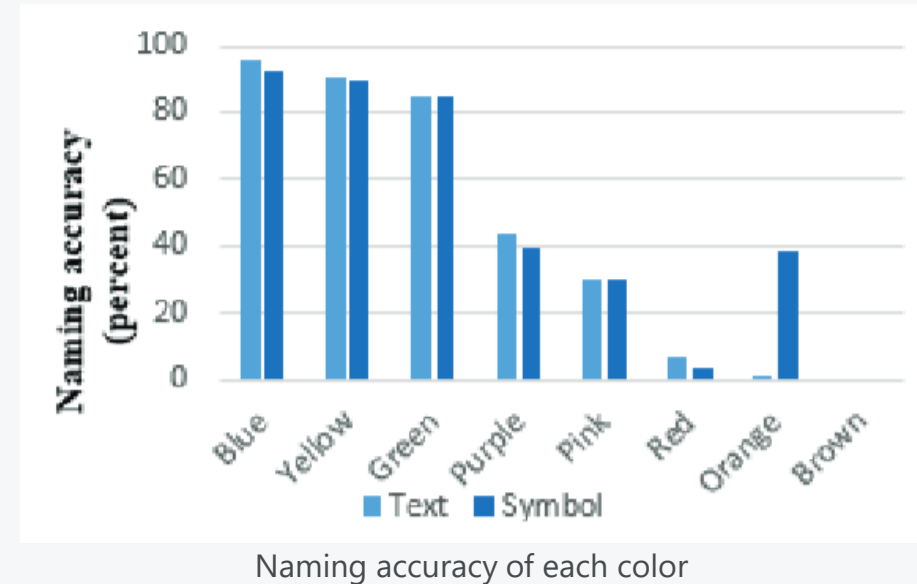
Results: Blue, Green, and Yellow are the Most Robust Colors

Color matching

- All color → Brighter shift
- Blue, Green, Yellow → Very little hue shift

Color naming

- Blue, Green, Yellow → High accuracy
- Brown → Poorly performed



Discussion: Color Robustness Depends on Chromaticity Consistency

Luminance dispersion: Little to no effect on user recognition

Chromaticity dispersion: Strong negative correlation with naming accuracy

Paper 3-3: Objective and Method

Objective:

Proposal of a Real-time Rendering Scheme to Enhance Color Contrast

Method:

Enhancement of Color Difference Using the Complementary Color

Color difference: Defined in CIE L*a*b space

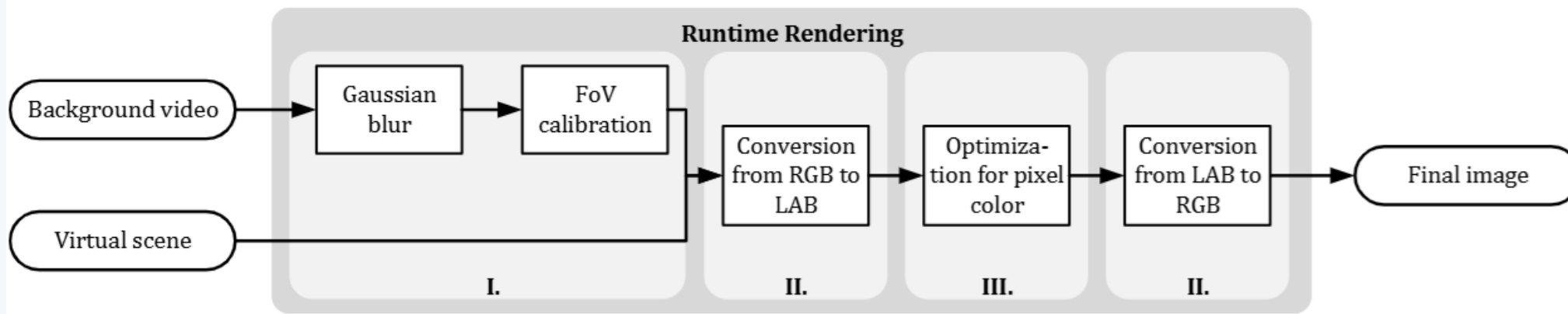
$$\Delta E_{ab}^*(x, y) = \sqrt{(L_x^* - L_y^*)^2 + (a_x^* - a_y^*)^2 + (b_x^* - b_y^*)^2}$$

Considering color semantics

4 constraints: Color Difference, Chroma, Luminance, Just Noticeable Difference (JND)

Paper 3-3: Algorithm

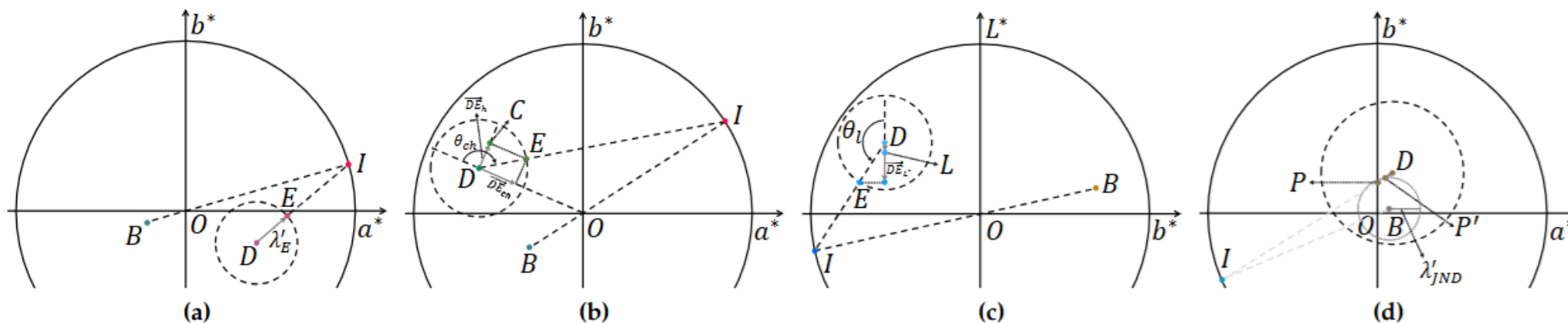
Preprocessing (blur, FoV Calibration) → Conversion(RGB to LAB) → Optimization



Optimization with constraints :

Algorithm Overview

(a)Color Difference (b)Chroma (c)Luminance (d)JND



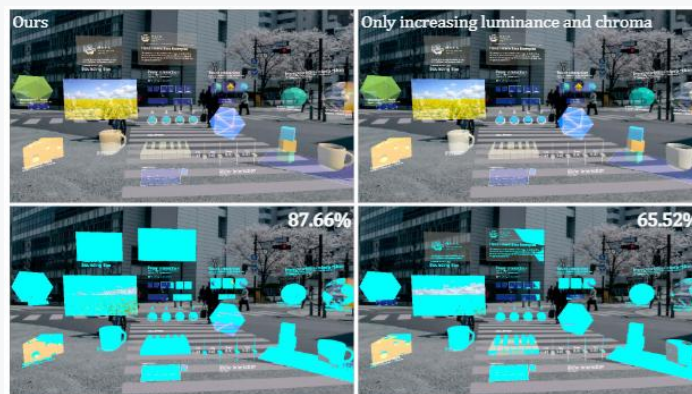
Algorithm on the 2D plane formed by two of the three axes of the LAB color space

Paper 3-3: Evaluation

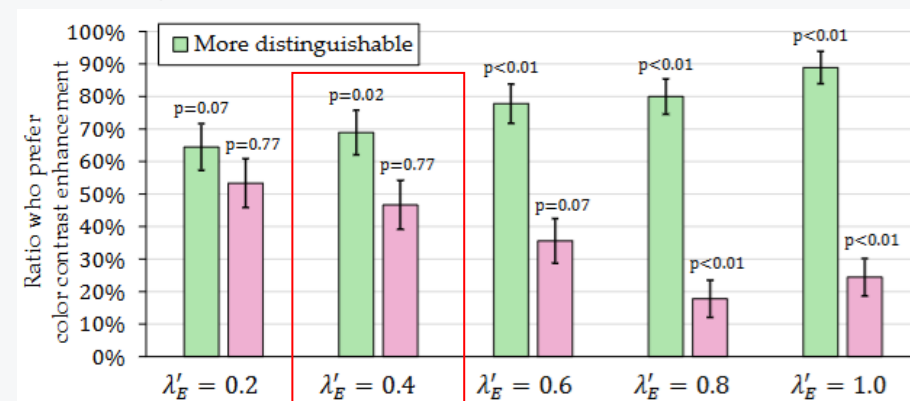
One Objective Evaluation and Three Subjective User Studies

Objective Evaluation in Simulation

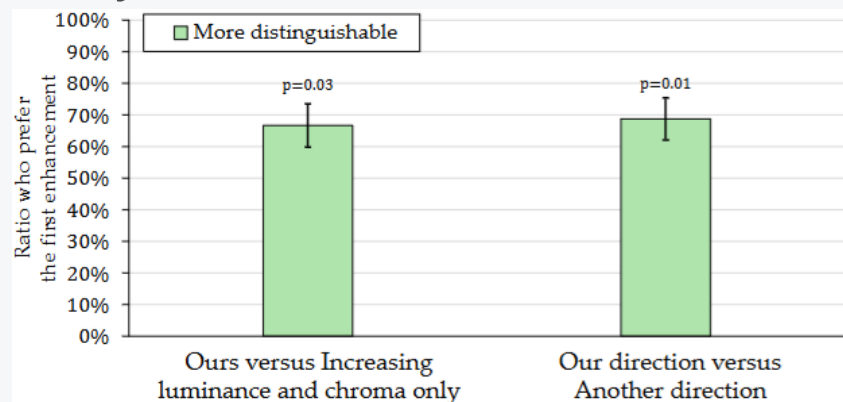
Cyan pixel
II
exceeds JND



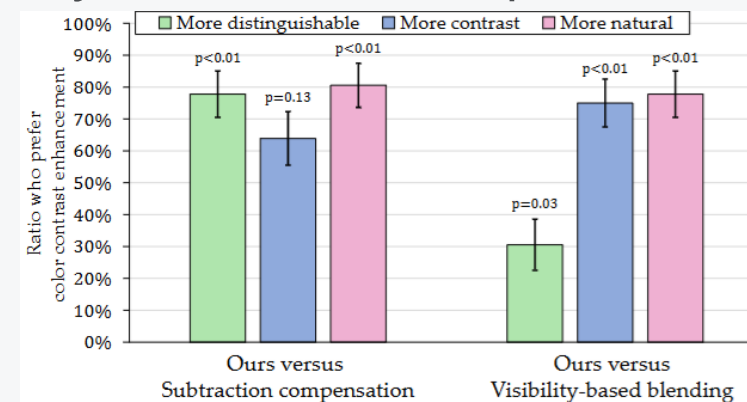
User Study1: Color Threshold Preference



User Study2: Hue Evaluation



User Study3: Method Comparison



Paper 3-3: Discussion and Conclusion

Discussion: Some Limitations of Proposed Method

- Need for adaptive threshold
- Ineffectiveness on achromatic
- Increase in power consumption
- Unsuitability for geometry

Conclusion: Summary of Contributions

- Real-time rendering scheme
- End-to-end algorithm
- Consistency with original color
- Enhance visibility

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Brightness and Visibility

Contrast Perception: 2 Color Channel and Luminance Channel

Brightness Factors

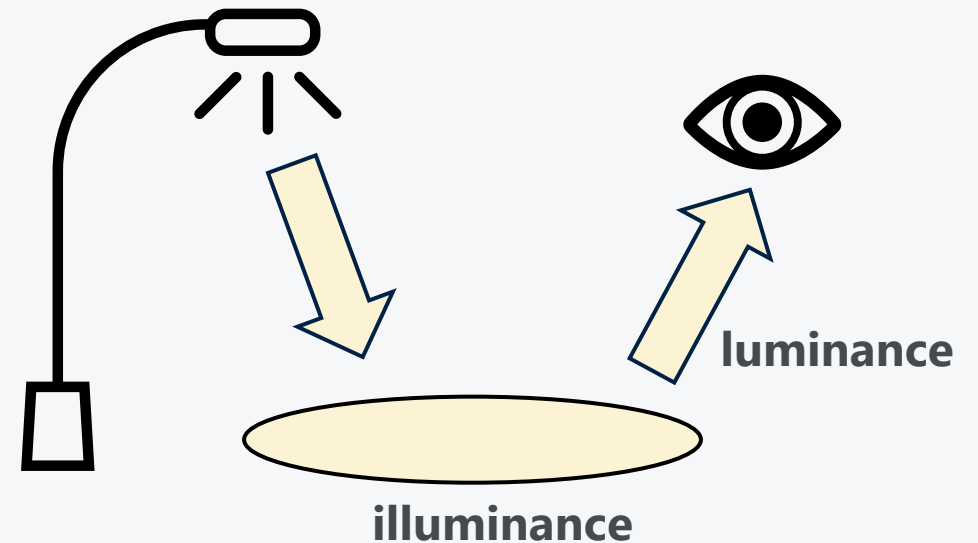
- Display luminance
- Ambient illuminance

Research outline

Paper4-1: Illuminance and Contrast

Paper4-2: Luminance and Layer Separation

Paper4-3: Luminance Prediction for Perceived Opacity

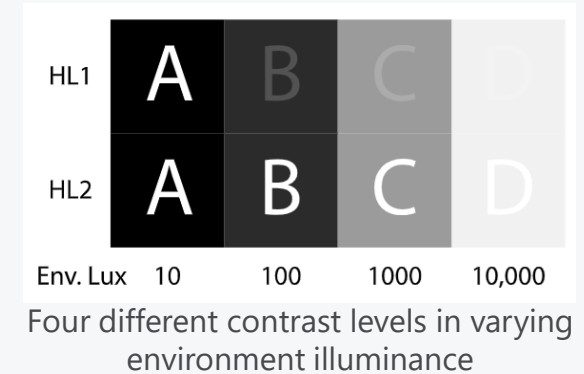


Paper4-1: Objective and Method

Objective:

Investigation of the Effect of Illuminance on Contrast in OST-AR

✕ Subjective user study ○ Engineering measurements



Method:

Measure Eye Illuminance under Various Ambient Illuminance Levels

Target illuminance levels: 0, 10, 100, 1,000, 10,000, and 20,000 lux

Sensor: Light meter

Device: Hololens1, Hololens2 (HL1, HL2)

Device mode: Without the HMD, Display Off, Display On (rendering black/white image)

Paper4-1: Results and Discussion

Reduction in Contrast as Environment Lighting Increased

Dim Lighting (<100 lux):

Contrast ≈ 1

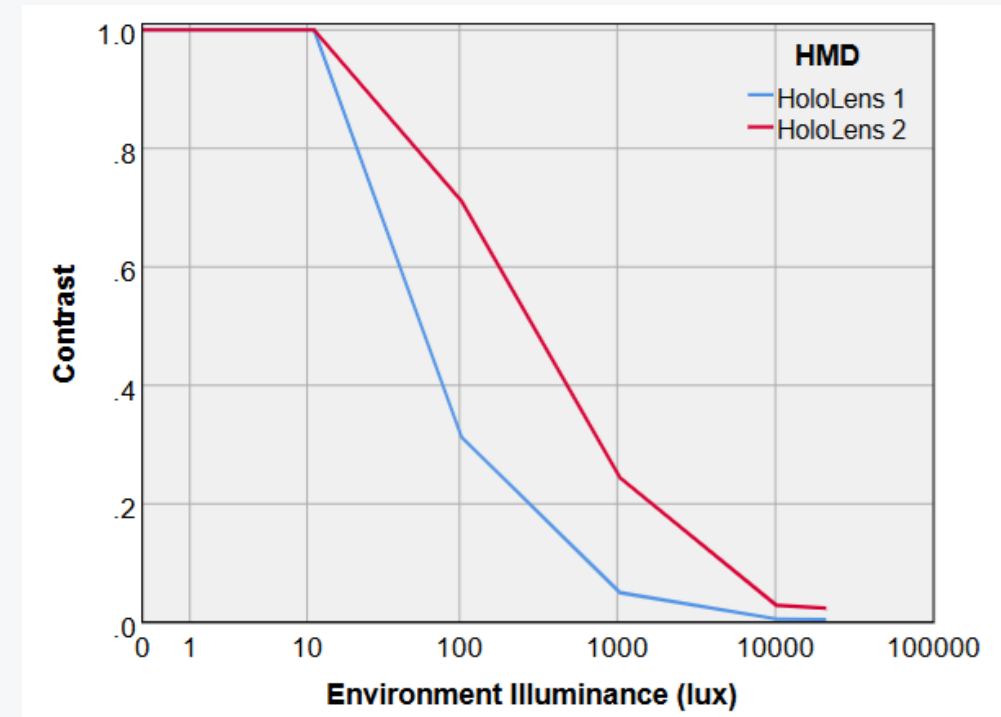
Mid-Range Lighting

100 lux: HL2 dropped to $\sim 70\%$

1,000 lux: HL2 dropped to $\sim 25\%$

Bright Outdoor Lighting (>10,000 lux)

Fell to 2–3% $\rightarrow$ Difficult to accurately distinguish virtual image



Depiction of the contrast ratios in relation to the environment illuminance levels for each HMD

Paper4-2: Objective and Method

Objective: Investigation of the Layer Separation Effect in OST-AR

Layer Separation Effect: Independent perception of BG and FG enabling constancy

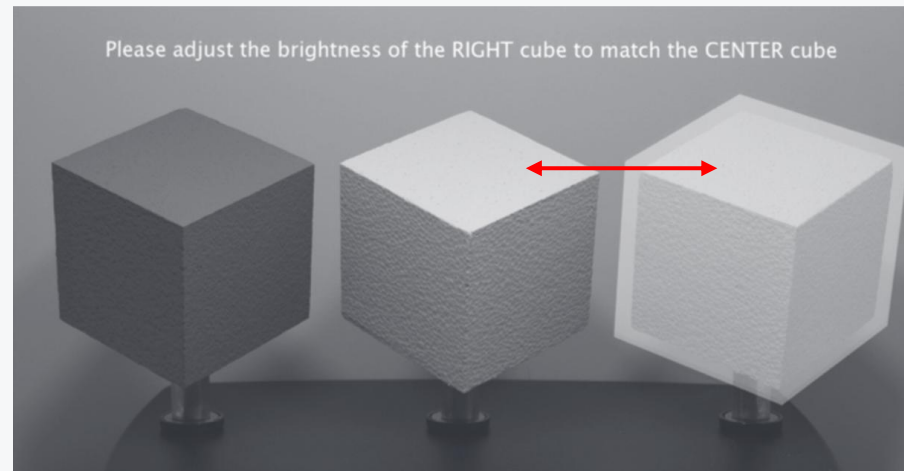
Method: Brightness-Matching Task

Adjusted the intensity of the AR overlay

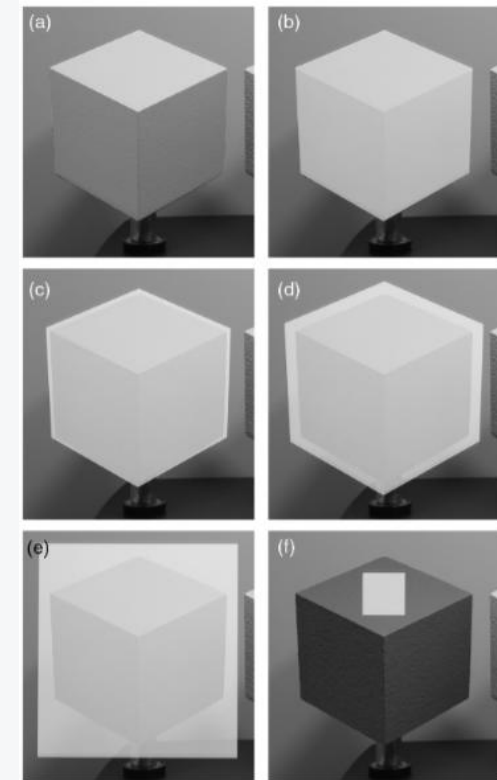
→ Match the brightness of the real-world cube

Stimuli:

6 types of AR overlay



Photograph of the observer's field of view showing the instructions



Overlays

Paper4-2: Results

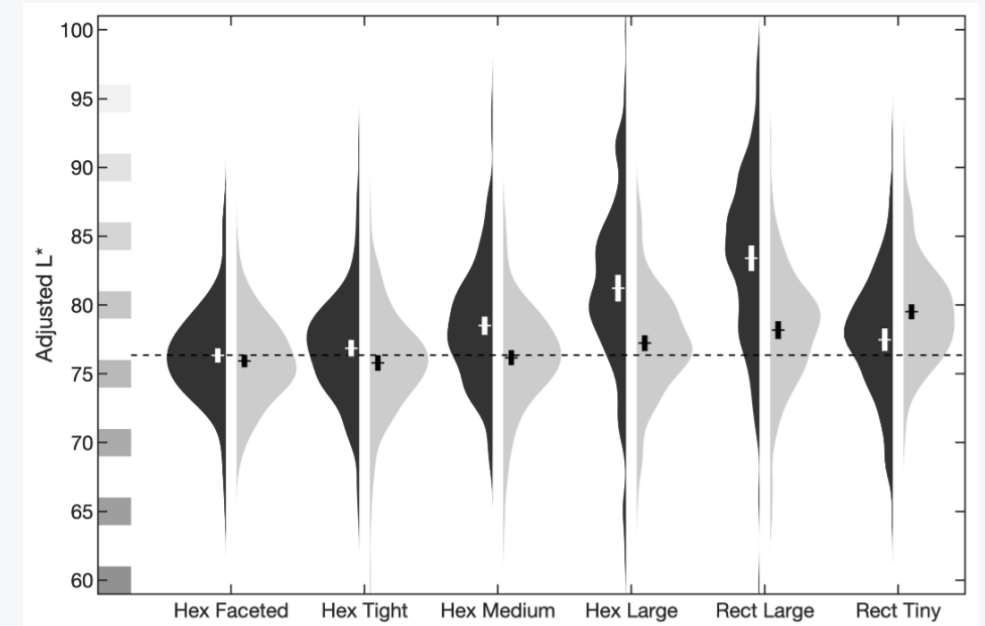
Results: Oversizing Increases Required Luminance

- Oversized overlays require more luminance
- Required luminance increases with overlay size

Hypothesis Supported

Oversized overlay

- More obviously perceived as a transparent layer
- Stronger foreground discounting
- More luminance is required



Overall results shown in a split-violin plot

Paper 4-3: Objective and Method

Objective: Prediction of Luminance Required for Opaque Perception

(1) Determine required luminance (2) Propose a prediction model

Method: Luminance Adjustment for Opaque Perception

Using custom HDR multi-focal display

$\sim 4000 \text{ cd/m}^2$

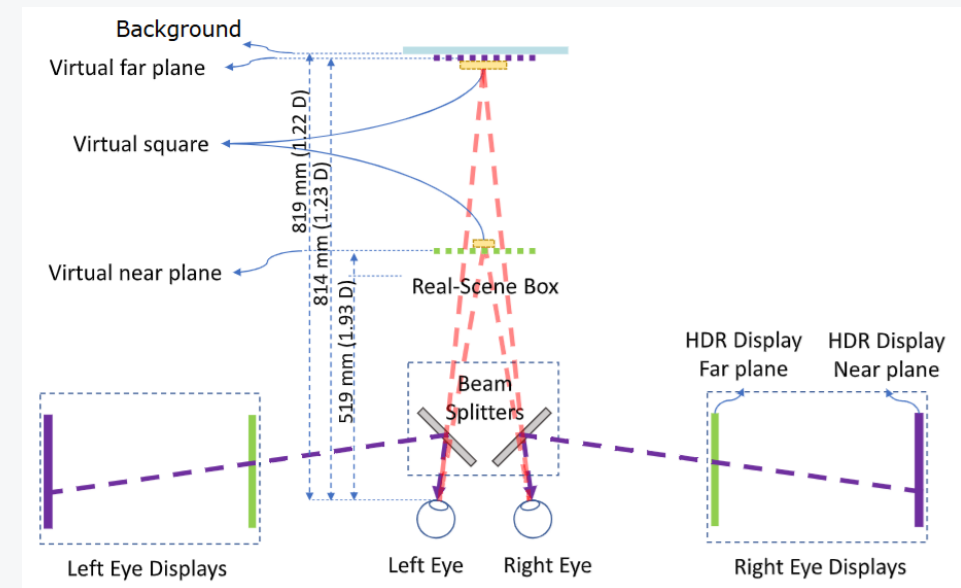
Stimuli

Background luminance: 2, 20 cd/m^2

Contrast of the virtual object: 0.1, 0.5

Frequency of the virtual object: 1, 6 cpd

Depth of the virtual object: 1.93, 1.23 D



The schematic of the high-dynamic-range stereo multifocal display

Paper 4-3: Results

Required Luminance: ~60x Background Level

Influencing factors

BG luminance: Dominant ($\sim 10x \rightarrow \sim 10x$ threshold)

Contrast: High contrast $\rightarrow$ less luminance

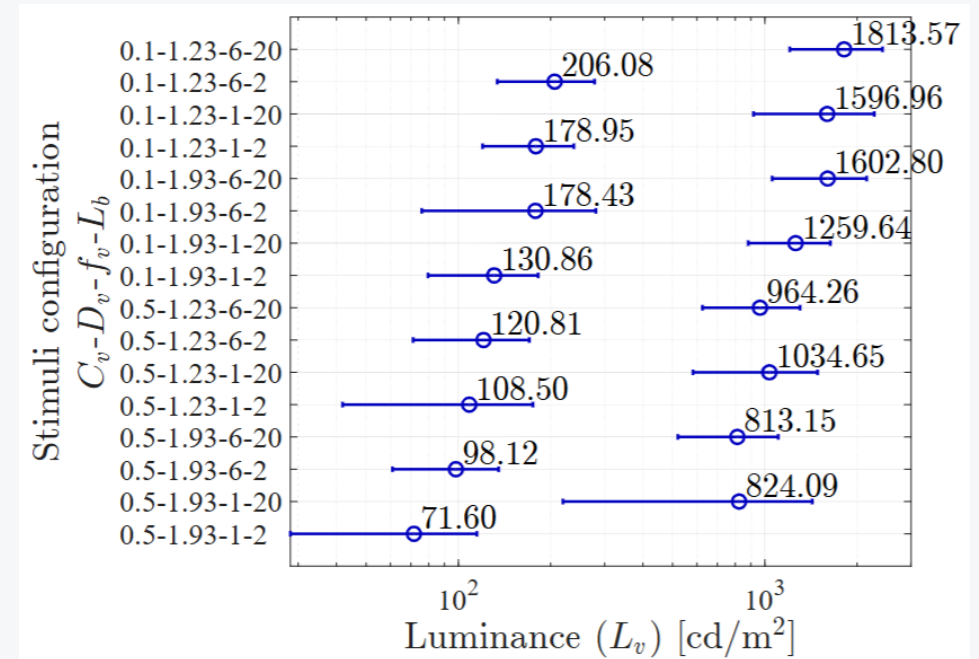
Depth: More blur $\rightarrow$ less luminance

Frequency: Negligible impact

Required luminance

Average: 60 times brighter than background

Mitigating Effect: reduce to 40 times (with high contrast and defocus)



The luminance threshold of opacity for the virtual object

Paper 4-3: Model

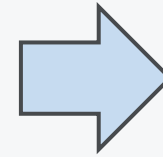
Model: Signal to Noise Ratio (Using Weber's Definition of Contrast)

Calculate BG contrast (weber's definition)

$$c_b = \frac{m_b}{L_b + L_t}$$

m_b : BG luminance amplitude
 L_b : Mean luminance of BG
 L_t : Mean luminance of virtual object)

Signal to noise ratio



$$S = \frac{c_b \cdot \text{MTF}(1)}{c_{det} + a c_t}$$

MTF: Modulation transfer function
 $S > 1$: Transparent
 $S \leq 1$: Opaque

Evaluation: Far More Accurate than Standard IQA Models

RMSE: 0.0346 log units **(8.29%)**

IQA Models(measuring transparent-to-opaque difference): 30%~

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Transparency and Visibility

Transparency = Unique Factor of OST-AR

Display mechanism

VST-AR = Image processing

OST-AR = Beam splitter → transparency

Research outline

Paper5-1: Perceived Transparency

Paper5-2: Perceived Transparency and illuminance

Paper5-3: Hardware-based Contrast Enhancement

Perceived Transparency

“The degree of visibility of an object through a medium”

(by International Commission on Illumination (CIE)[1])

Paper5-1: Objective and Method

Objective:

Investigation of Background Effects on Perceived Transparency

Background effect → luminance, contrast, wave form

Method: Psychophysical Scaling under Varying Background Conditions

Direct scaling: 0 ~ 1

Stimuli

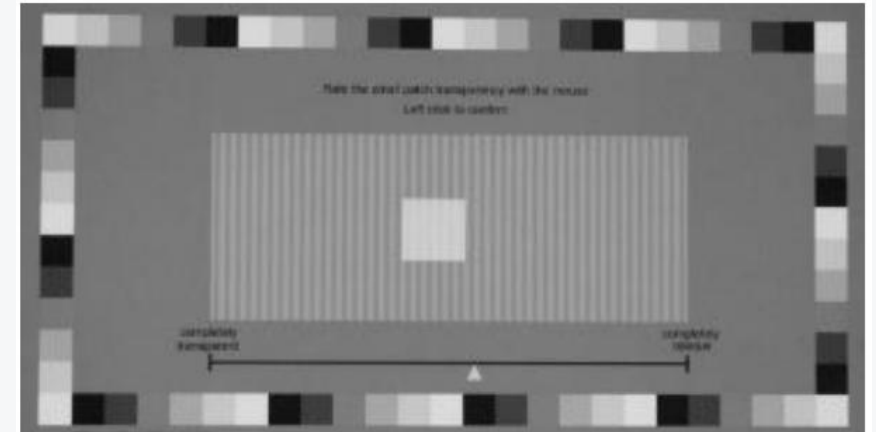
Luminance: low, mid, high (LL,ML,HL)

Contrast: low, mid, high, very high (LC,MC,HC,HHC)

Wave form: sine, square

Relative Contrast: 10 steps

Types of overlay: OST-AR, Simulated



Photograph of example experiment interfaces

Paper5-1: Results

Results: Background Contrast Has a Significant Effect

OST-AR vs Simulation

Fundamental differences → Limitations of simulation

Effect of factors

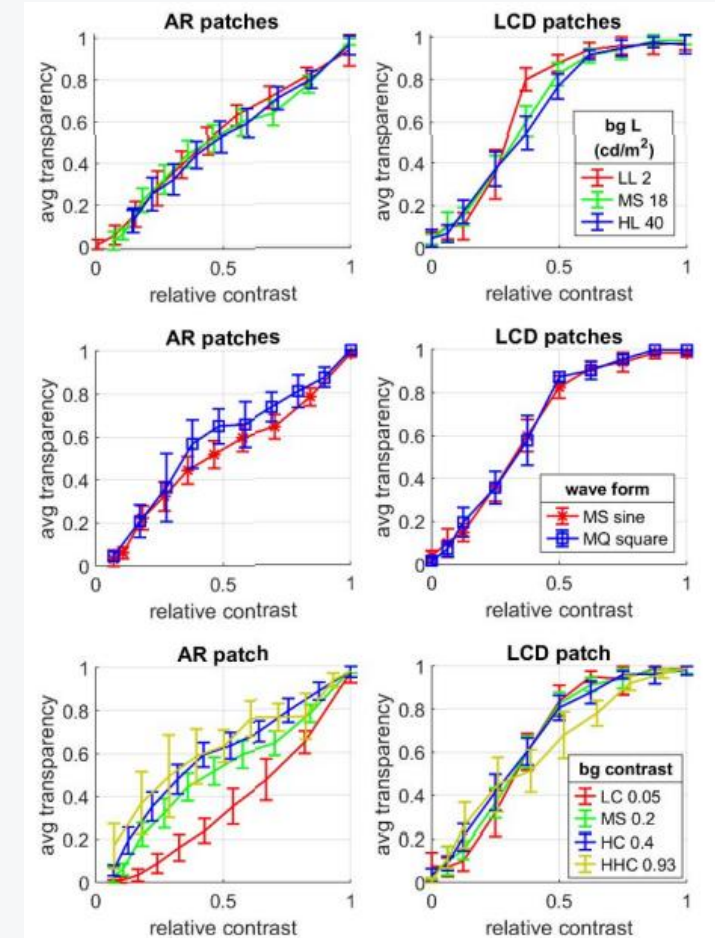
High contrast → Increases perceived transparency

Luminance, waveform → No significant effect

Modeling: Based on Michelson Contrast

$$T = 1 - e^{-\left(2.5 \frac{C}{C_b}\right)^{1.75}} - 0.05$$

High accuracy:
Correlation ≈ 0.99



Top row: effect of background average luminance

Middle row: effect of background wave form

Bottom row: effect of background contrast

Paper5-2: Objective and Method

Objective:

Investigation of Ambient Illuminance on Perceived Transparency

Method: Psychophysical scaling under Varying Ambient Illuminance

Direct scaling: 0% ~ 100% (Step size: 1%)

Stimuli

Illuminance: low, high (10~20, 400~500 lux)

True transparency level: 10% ~ 100% (Step size: 10%)



Experimental scene

Paper5-2: Results and Discussion

Results: Underestimation of Transparency

- (1) Increase physical transparency → Underestimate perceptual transparency
- (2) High illuminance → More underestimate

Discussion: (2) ← Reduction of Relative Brightness of Overlay?

This mean: High illuminance → More opaque perception (→ High visibility)



Paper3-1: High illuminance → Low contrast ratio → Low visibility

Propose another Hypothesis (consistent with Paper5-1)

Bright environment → ambient light illuminates the background display
→ Reduction in background contrast → reduction in background contrast

Paper5-3: Objective and Method

Objective:

Improvement of the Visibility of OST-HMD via Hardware

Limitations of Conventional Methods: Image quality degradation, Excessively bulky

Method: Adopts a GHLC cell to Adjust Transmittance via Applied Voltage

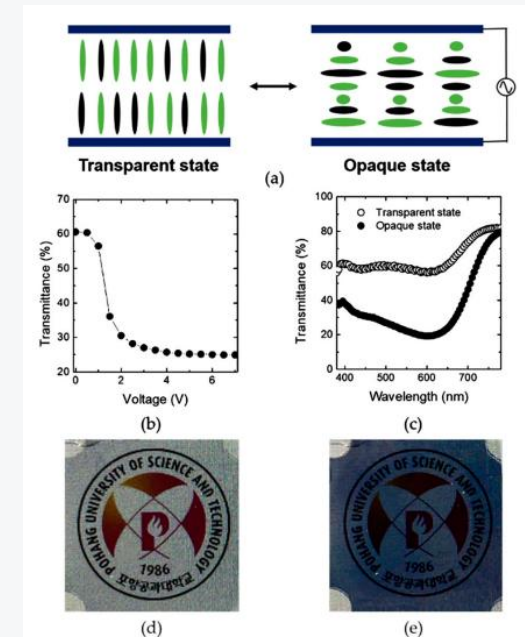
GHLC cell

Host: Liquid Crystal (LC) molecules (chiral dopants)

Guest: dichroic dye molecules

Initial (Vertically aligned) → Transparent

Voltage applied (Twisted structure) → Opaque



The GHLC cell using inverse-STN mode

Paper5-3: Experiment

Luminance and Color Measurement in Varying Ambient Illuminance

Environments (Illuminance)

Dark room (**0 lux**)

Office environment (**600 lux**)

Outdoor-equivalent
bright environment (**6000 lux**)

Evaluation Metrics

Ambient Contrast Ratio (ACR)

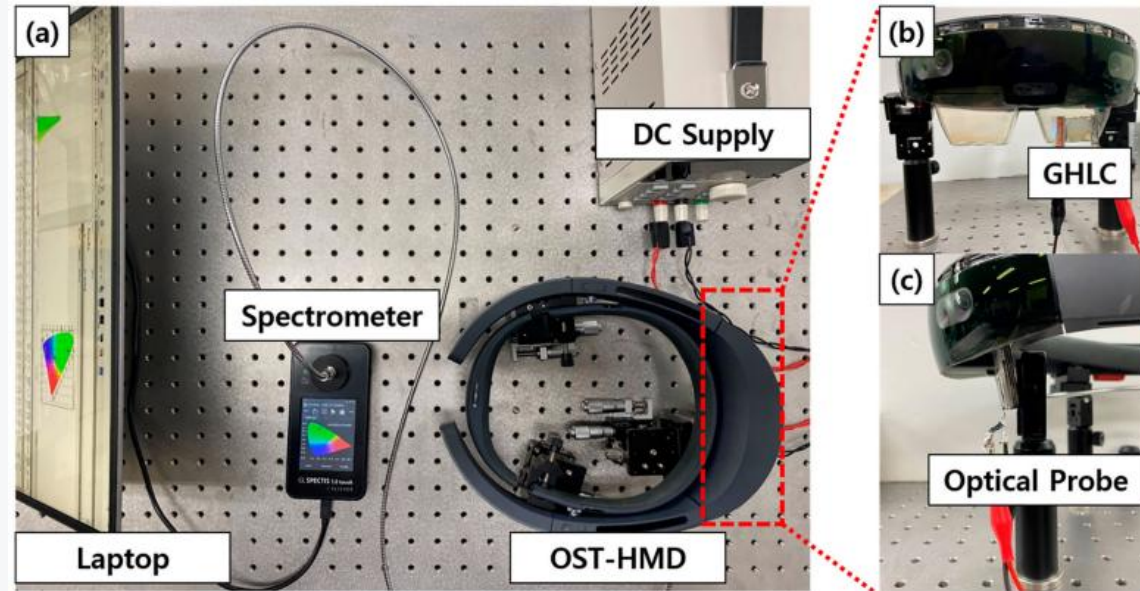
$$ACR = \frac{L_W + L_B + L_R}{L_K + L_B + L_R}$$

L_W :Luminance of the white state

L_K : Luminance of the black state

L_B : Background luminance

L_R : Luminance due to reflection or glare



Experimental setup

Paper5-3: Results and Conclusion

Results: Improvement of Contrast Ratio and Color Gamut

Optical characteristics								
Ambient light condition		Dark room (0 lux)		Office (600 lux)		Outdoor (6000 lux)		
GHLC State		None	None	Off	On	None	Off	On
Luminance [cd/m^2]	White	230.25	454.57	318.15	261.36	731.62	530.49	356.71
	Black	0.68	68.67	42.09	8.41	487.54	503.82	154.10
Ambient contrast ratio		338.6	6.62	7.56	31.07	1.5	1.05	2.31
Color coordinate (x,y)	Red	(0.6517, 0.3137)	(0.5032, 0.3463)	(0.5066, 0.3413)	(0.5610, 0.3197)	(0.3714, 0.3669)	(0.4079, 0.3565)	(0.4315, 0.3256)
	Green	(0.2054, 0.6999)	(0.2758, 0.5734)	(0.2701, 0.5633)	(0.2127, 0.6810)	(0.3287, 0.4142)	(0.3334, 0.3922)	(0.2777, 0.4477)
	Blue	(0.2176, 0.1494)	(0.2909, 0.2877)	(0.2894, 0.3008)	(0.2251, 0.1602)	(0.3391, 0.3623)	(0.3257, 0.3395)	(0.2774, 0.2683)
Color gamut [%]		76.16	19.45	18.27	55.91	0.54	1.33	8.73

Office environment

ACR: 6.62 → 31.37

Color Gamut: Improved x2.87

Outdoor environment

ACR: 1.5 → 2.31

Color Gamut: Improved x16

Conclusion: Substantially Increased the ACR and Color Gamut

Detachable form → Can be applied to existing OST-HMDs

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Visibility Predictor

Lacking comprehensive approach for OST-AR visibility

Paper 3-3, 4-3,5-3 → Addressing individually

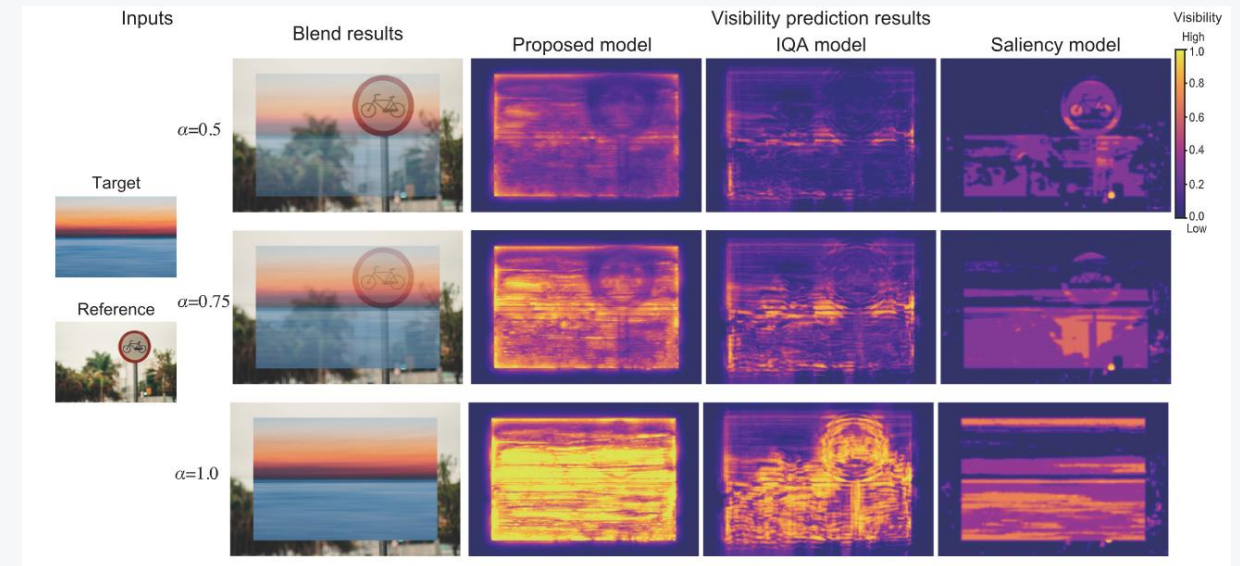
Promising model for comprehensive prediction ↓

Content-adaptive Visibility Predictor proposed by Fukiage et al.

Objective: Predict visibility of target images in α -blended image

Limitation of Conventional IQA Models

Content Independence, background effect

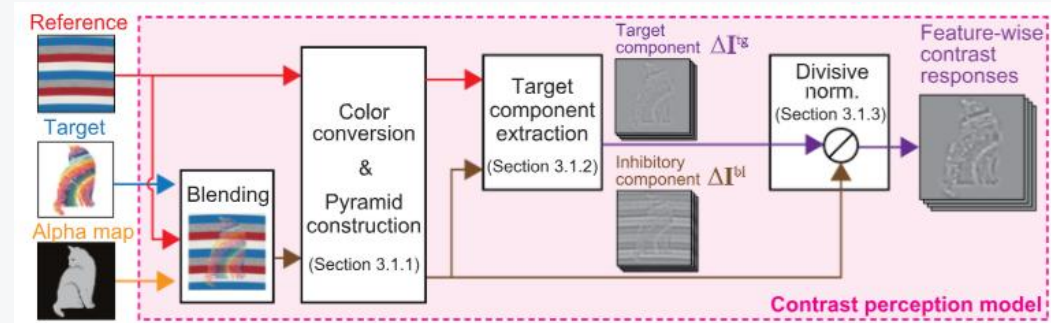


Comparison of supra-threshold visibility maps

Visibility Predictor: Key Features and Performance

Three Key Features

1. Contrast Perception Mechanism: Contrast masking, HVS visibility perception
2. Target Component Extraction: Separate background and target
3. Content-Adaptive Feature Aggregation: Target-dependent weight calculation



Flow chart of proposed model

Performance: Calibrated via Psychophysical Visibility Matching Task

Visibility=How well you can see features contained in the image

Achieved higher prediction accuracy than existing IQA metrics

Visibility Predictor: Challenges to Apply OST-AR

Potential for OST-AR



Implementation Challenges

Model input

SDR (α -blending) vs HDR (OST-AR)

Visual Mechanisms

Binocular combination, Defocus blur, Layer separation

Model calibration

Recalibrate with OST-AR visibility data, via psychophysical experiment

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Conclusion

Perceptual Visibility on OST-AR → Prediction, Enhancement, IQA, UX, Safety

	Color	Brightness	Transparency
Basic	Color blending → Hue Shift	High illuminance → Reduce Contrast	High contrast → Increases perceived transparency
	Robust colors → Blue, Green, and Yellow	Required Luminance → Increases due to Oversizing	Underestimation of Transparency
Applied	Enhancement of Color Difference → Enhance visibility	Signal to noise ratio → Predict the luminance for perceived opaque	GHLC cell → Improve Contrast Ratio and Color Gamut

➡ **Many other researches, results, approaches**

Visibility Predictor → Many challenges to apply OST-AR